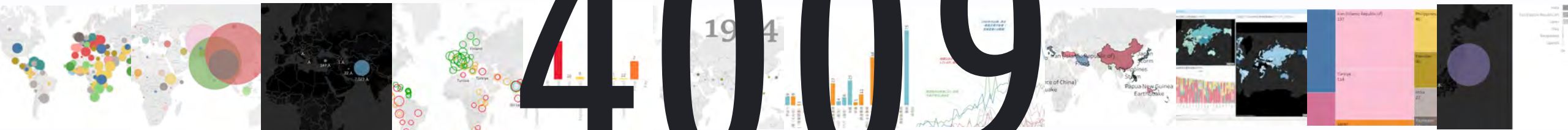


Gallery walk

4009



Mapping Events/ Mapping Social Media

Participatory Digital Humanities

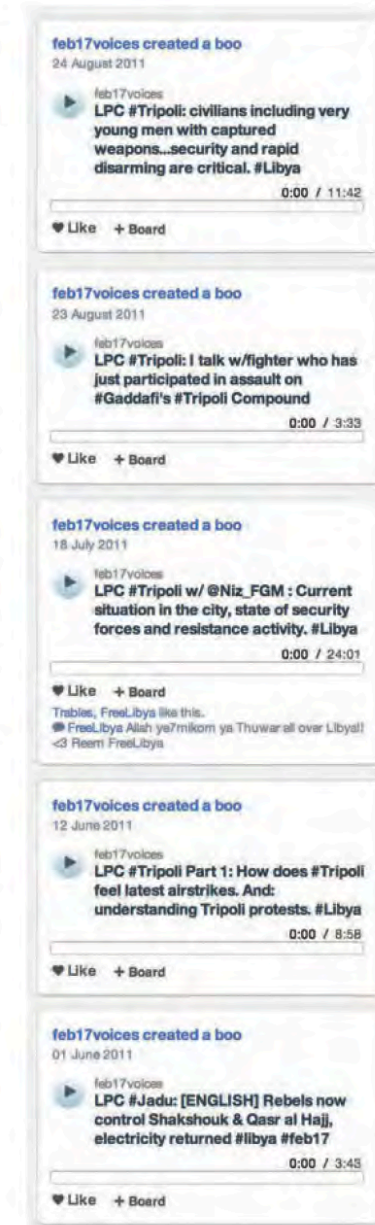
(TP) If there is a utopian idea at the core of the digital humanities, it would have to be the notion of “participation without condition”—that is to say, participation in the creation of the cultural record of humanity regardless of facticity and, hence, not dependent upon or restricted by race, gender, class, religion, ethnicity, nationality, age, language, access to technology, or education. Participation



begun to render the walls of the university porous by engaging with significantly broader publics in the design, creation, and dissemination of knowledge. By conceiving of scholarship in ways that foundationally involve community partners, cultural institutions, the private sector, non-profits, government agencies, and ever-broader slices of the general public, digital humanities expands both the notion of scholarship and the public sphere in order to create new sites and nodes of engagement, documentation, and collaboration. With such an expanded definition of scholarship, digital humanists are able to place questions of social justice and civic engagement, for example, front-and-center; they are able to revitalize the cultural record in ways that involve citizens in the academic enterprise and bring the academy into the expanded public sphere.

The result is a form of scholarship that is, by definition, translational and applied: it applies the knowledge and methods of the humanities to pose new questions, to design new possibilities, and to create citizen-scholars who value the complexity, ambiguity, and differences that comprise our cultural record as a species.

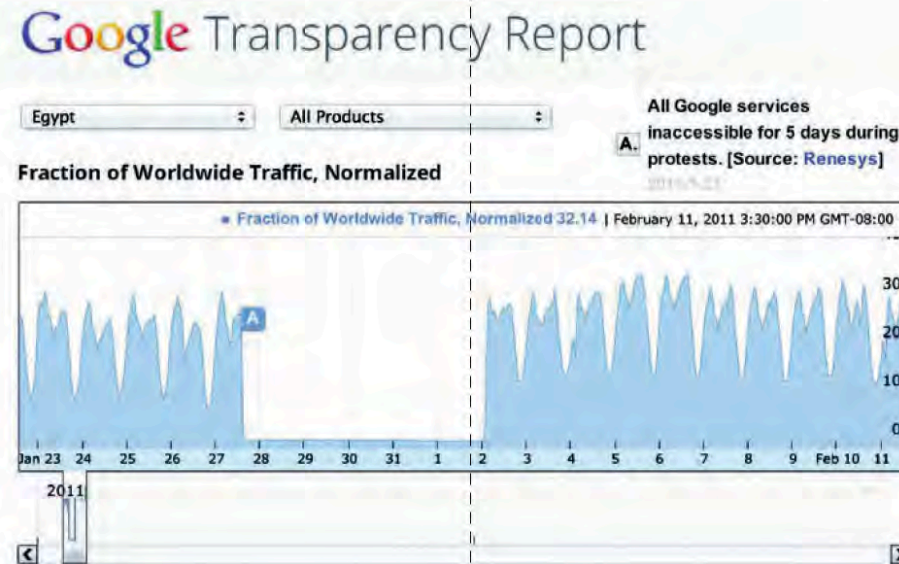
The "Voices of January 25th" and "Voices of February 17th" documentary projects are compelling examples of how



social technologies like Twitter can be used to give voice to people who were silenced in the 2011 revolutions in Egypt and Libya. Started by John Scott-Railton, then a graduate student at UCLA, the projects used Twitter to disseminate suppressed messages from dissidents to the world. Scott-Railton began the Voices of January 25th project when Egypt effectively “turned off” the Internet between January 28 and February 2, 2011.

Relying on cell phones and later landlines, Scott-Railton began calling friends in Egypt who knew protesters and could provide highly localized, accurate accounts of what was happening on the ground. He assembled a network of trusted informants who agreed to have their phone calls recorded and their voices published to the world on AudioBoo, an audio hosting service. Scott-Railton, then, posted messages to Twitter, often with links to audio files and other media reports that would help the world “see” and “hear” what was going on in Egypt in real-time. In effect, the digital portal became a global public sphere, however fragile and endangered, that was fundamentally linked to the deeply embodied and precisely located events on the ground. The voices are now part of a living web archive and documentary memorial.⁷⁰

While the role of social media has been feverishly de-



bated in fomenting, planning, and sustaining revolutions since Twitter was first hailed—somewhat exaggeratedly—as a revolutionary technology in Moldova in 2009 and YouTube became a people’s archive for election protests in Tehran during the summer of that same year,⁷¹ it seems incontestable that the “public images” produced by broadcast media (often singular, uni-directional, and hierarchical) are being supplanted by decentralized, multi-directional “public utterances” that are changing the way in which events unfold, become

represented, and are disseminated (almost instantaneously) on a global scale. Two decades ago, Paul Virilio thought that the physical space of the public sphere had become replaced by the media of the “public image”; but now we are seeing the reassertion of the public sphere through radically non-Cartesian geographies enabled by what I earlier termed the web’s “contiguity of the non-contiguous.”⁷²

What this means is a massive contraction and alignment of the event (an embodied and location-specific phenomenon), the representation of the event (through Twitter messages, blogs, video, and photographs captured and uploaded from mobile phones, and so forth), and the dissemination of the representation (through web-based social networks and information channels). But more than just an alignment of event, repre-

content for the “Digital Archive of Japan’s 2011 Disasters,” and with Niigata University, the team has developed a web interface and visualization tool for mapping real-time radiation levels. These projects are discussed more extensively below.

On the Event: HyperCities Now (Egypt, Libya, Japan)

Every event is singular but enmeshed in an infinite number of other events (space).

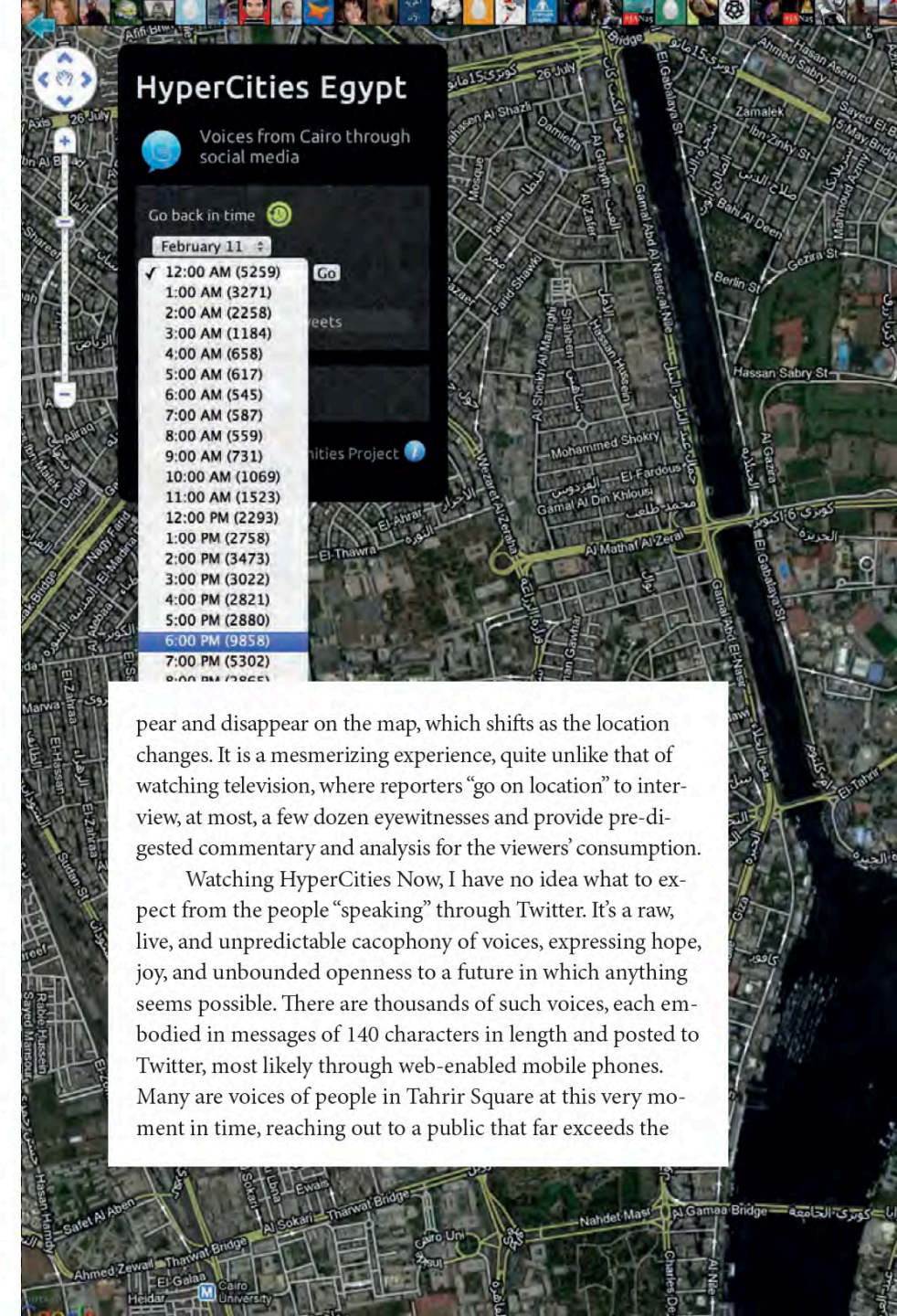
The event, its representation, and the dissemination of the representation coalesce in feedback loops in which the event is never the same.

Events happen on bodies and are expressed in language and through visual media.

Events are processes of becoming with multiple temporalities.

The event has no end (time).

(TP) A message flashes on the screen. It is by a user who goes by the handle PortSa3eedy: “Chants in Tahrir: ‘Muslims, Christians, We are One.’ This is the rebirth of a country, with a feeling that anything is possible #Egypt.” The time is February 11, 2011, 07:22:55 PM in Egypt. It’s about an hour after Mubarak delivered his resignation speech. The location returned is city-center Cairo. The Twitter message stays on the screen for about four seconds, before another message flashes on the screen. I watch for hours, as thousands of messages ap-



pear and disappear on the map, which shifts as the location changes. It is a mesmerizing experience, quite unlike that of watching television, where reporters “go on location” to interview, at most, a few dozen eyewitnesses and provide pre-digested commentary and analysis for the viewers’ consumption.

Watching HyperCities Now, I have no idea what to expect from the people “speaking” through Twitter. It’s a raw, live, and unpredictable cacophony of voices, expressing hope, joy, and unbounded openness to a future in which anything seems possible. There are thousands of such voices, each embodied in messages of 140 characters in length and posted to Twitter, most likely through web-enabled mobile phones. Many are voices of people in Tahrir Square at this very moment in time, reaching out to a public that far exceeds the

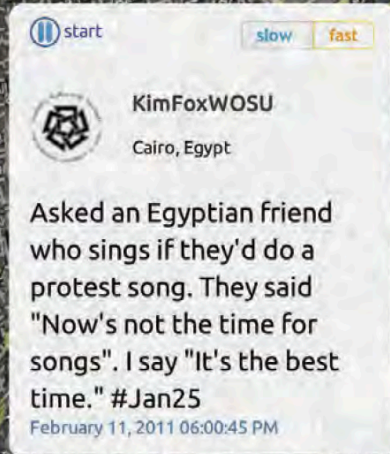
apprehension of any human listener in this place, at this time. On Twitter, the singularity of the voice is transported, refracted, and multiplied. It becomes part of a collective event and attaches itself to anyone ready to listen, anyone attuned to its message, anyone standing open.

Every tweet that flashes up on the screen is a pure event. It cannot be predicted and, in a sense, breaks free of time; yet when it happens, the tweet is utterly bound to a “kernel of time” (*Zeitkern*). In the same way, the tweet cannot be specified precisely on a map ahead of time, but when it happens, it is utterly “incarnated,” the action, thought, and message of a person in a particular place. As a viewer, I cannot know when a tweet will come, from where it will come, who will send it, and what it will say. In this sense, it is a flash, a moment of truth of someone sending a message into the ether.

Only after the event can we begin to make sense of it as an event, by putting it in a timeline, by considering it as part of a network, by analyzing it for meaning, by investigating it as a part of a whole. This is the realm of hermeneutics, computational analysis, statistics, historical interpretation and explication. It happens after the fact.

There are three kinds of temporality at work in Hyper-Cities Now:

The first is the time of the flash, the Aion as pure event, the Twitter message that appears on the map at a singular moment in time. It is an absolute surprise, utterly unexpected and unpredictable, even though its conditions of possibility can be anticipated and its context understood. This is the “now-time,” the flash Walter Benjamin called the *Jetztzeit*, an instant or eruption that distills or contracts the present and harbors the notion of the revolutionary, the messianic, the possibility of a future that cannot be derived from the stock



of historical experiences. Anything is possible: justice, democracy, freedom, but also the violent backlash, the disaster, the catastrophic.

The second kind of temporality is that harbored in the event as network, when it becomes an event with hopes and expectations of a future that is not yet and memories and traces of a past that is no longer. It is reflected in the viewer's perception of the flash itself: when we—those who are corporeally “uninvolved” in the event, those who are “far away” as spectators—perceive the Aion, the event has already happened (it is no longer) and another event is about to happen (it is not yet). In both, there is a deep contingency. Recalling Aristotle's definition, the contingent is that which is neither necessary nor impossible. It didn't have to happen this way, and it could have been otherwise.



The third kind of temporality is that of **Chronos**, the historical time in which events occur, the timeline of events within chronological time, which help us make sense of events for analysis, comparison, and hermeneutical interpretation. It is the archive of tweets, searchable by hour and day, viewable by minute and second, in relationship to every other tweet by any other user who also used certain hashtags to describe, document, or otherwise witness events on the ground, in Egypt, during the revolution that began on January 25, 2011.

Start-dates are comparatively straightforward to assign, especially when world-historical events have demarcated triggers, such as January 25, 2011, or March 11, 2011, the day of the earthquake, which struck at 2:46 PM, and tsunami that enveloped Japan shortly after (3:38 PM). Such events cannot be readily derived from prior history (although their meanings and understandings will inherit much from the stock of historical knowledge and experience). They could not have

been predicted to occur *on that day, at that time*. For that reason, world-historical events are first of all events that cannot yet be comprehended. They don't yet have full meaning because they are eruptions, fissures that are violently torn from the everyday. For someone "inside" the event (by which I mean someone whose body is affected by the event, who is physically proximate to the event and is, therefore, touched by it in a way that is different from someone who experiences the event through mediations and representations alone), it is impossible to know the extension and duration of the event, its scope and scale, its impact and significance, its resonance and meaning. The event is happening, and only later, with the passage of time and the (comparative) clarity of distance, can eyewitnesses make sense of the event by perceiving its contours. For someone "outside" the event (whose body is ostensibly safe and untouched), the event assumes certain contours right away, even if these contours are erroneous or incomplete, through the media in which it is represented and disseminated (on television, on radio, on the web, via Twitter, and so forth). It is foremost a spectatorial event, which also has the effect of placing the viewer in a precarious ethical relationship to the suffering of those "inside" the event.

I watched the Twitter streams flash on the map from the safety of my desk in my office at UCLA, placing me in a questionable ethical position of a privileged observer of events that I took no part in. For the person sending a message, the vulnerabilities cannot be overstated: who is to know if the message will be tracked and put into another kind of database, not one that archives the voices of history but one that aims to squelch those very voices? Who is to know if data will be harvested by others in order to more fully exert control and homogenize history? I am only an ethical spectator—a witness to the witnesses—if I am willing to be open to

through its API. We privileged some hashtags (#egypt, #jan25, and #tahrir) over others. And, of course, Twitter itself is a self-selecting community, primarily youth, many from educated and wealthier backgrounds, who had access to this technology.⁷⁹

Egypt: A Global View

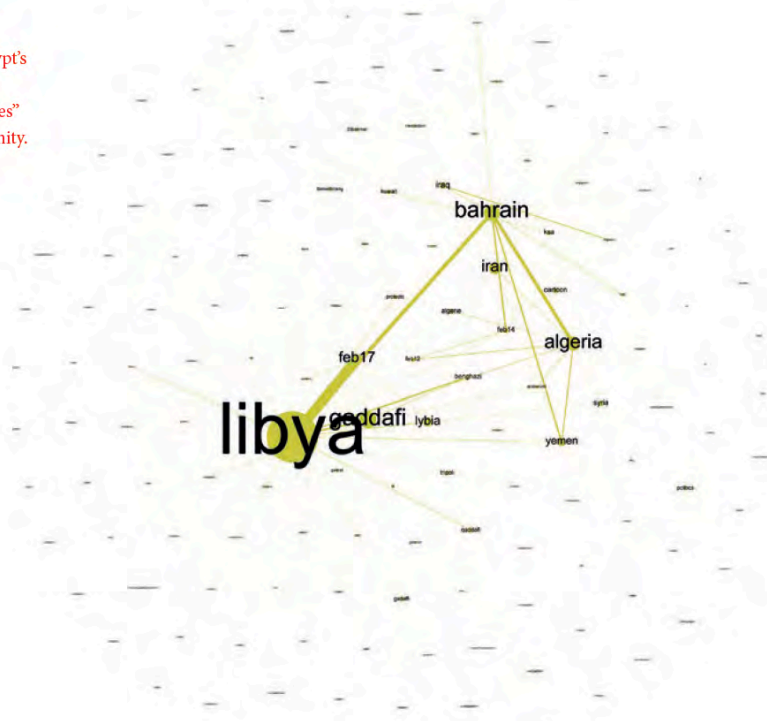
(DS) What allows us to call the Egyptian revolution an "event"? We cannot have faith in a definition of the event in the singular, a suspect concept once we begin to consider events in layered, multi-functional contexts. Instead, what if we imagine an event as a network, and try to visualize it as such? Networks invite us to imagine peers working together to pass on information, without official communication channels. Network analysis is an apt method for analyzing events represented and enacted through social media. Ideas (whose signifiers are hashtags) are simply nodes, manifested only by the individual acts of communication users send to each other. The diversity of discourse bears witness to the complexity of an event and the number of people and messages involved; in theory, networks show many revolutions rather than one revolution.

Figure 21 shows all the hashtags used in tweets gathered by "HyperCities Now" during the early period of the Egyptian revolution⁸⁰; each node is a hashtag and a line between two nodes represents the messages that contained those two hashtags.⁸¹ This is the global level of an event on Twitter. Nodes' size reflects how frequently they were mentioned. We recorded these tweets by tracking three hashtags: #jan25, #egypt, and #tahrir, but 3,520 other hashtags occur with

(21) Network graph of hashtags used during the Egyptian Revolution.



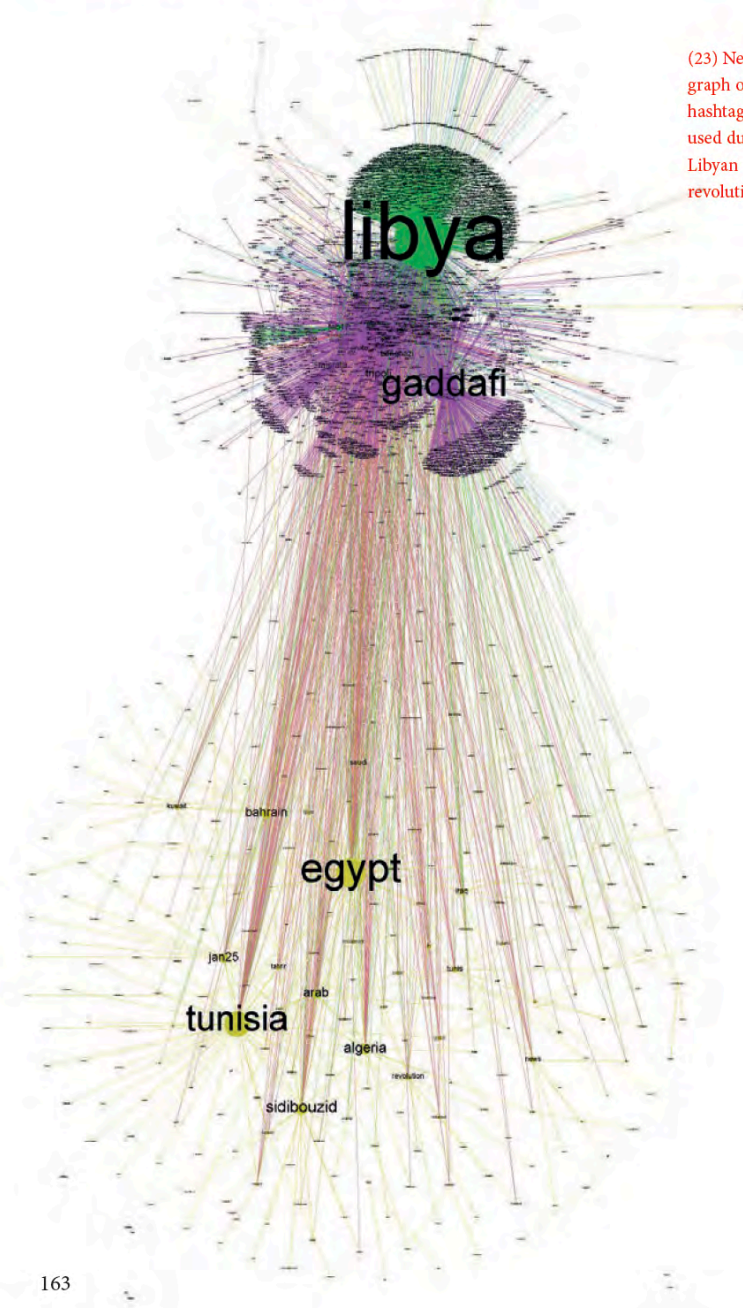
(22) Egypt's
"Other
Countries"
community.



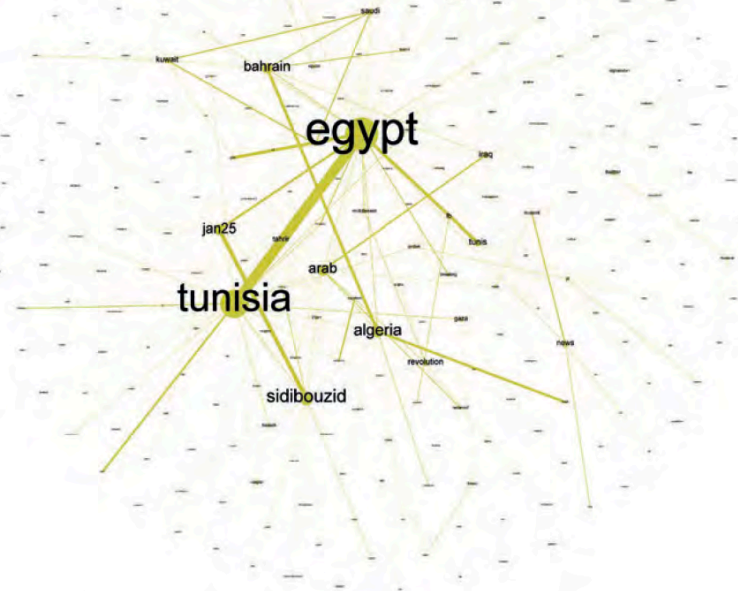
these three hashtags. This network looks like a galaxy with the three tags we tracked as the suns that exert their pull on the other stars.

Alongside its suns, this galaxy has solar systems. In network theory, these are called "communities," sets of nodes more strongly connected to each other than to other nodes.⁸² Colors highlight different communities. The red nodes are those most strongly connected to the three central nodes. This is the "Egyptian revolution" community. Not surprisingly, it is the largest. There is another smaller one, centered on #mubarak. There are, in fact, 200

(23) Network
graph of
hashtags
used during
Libyan
revolution.



(24) Libya's
"Other
Countries"
community

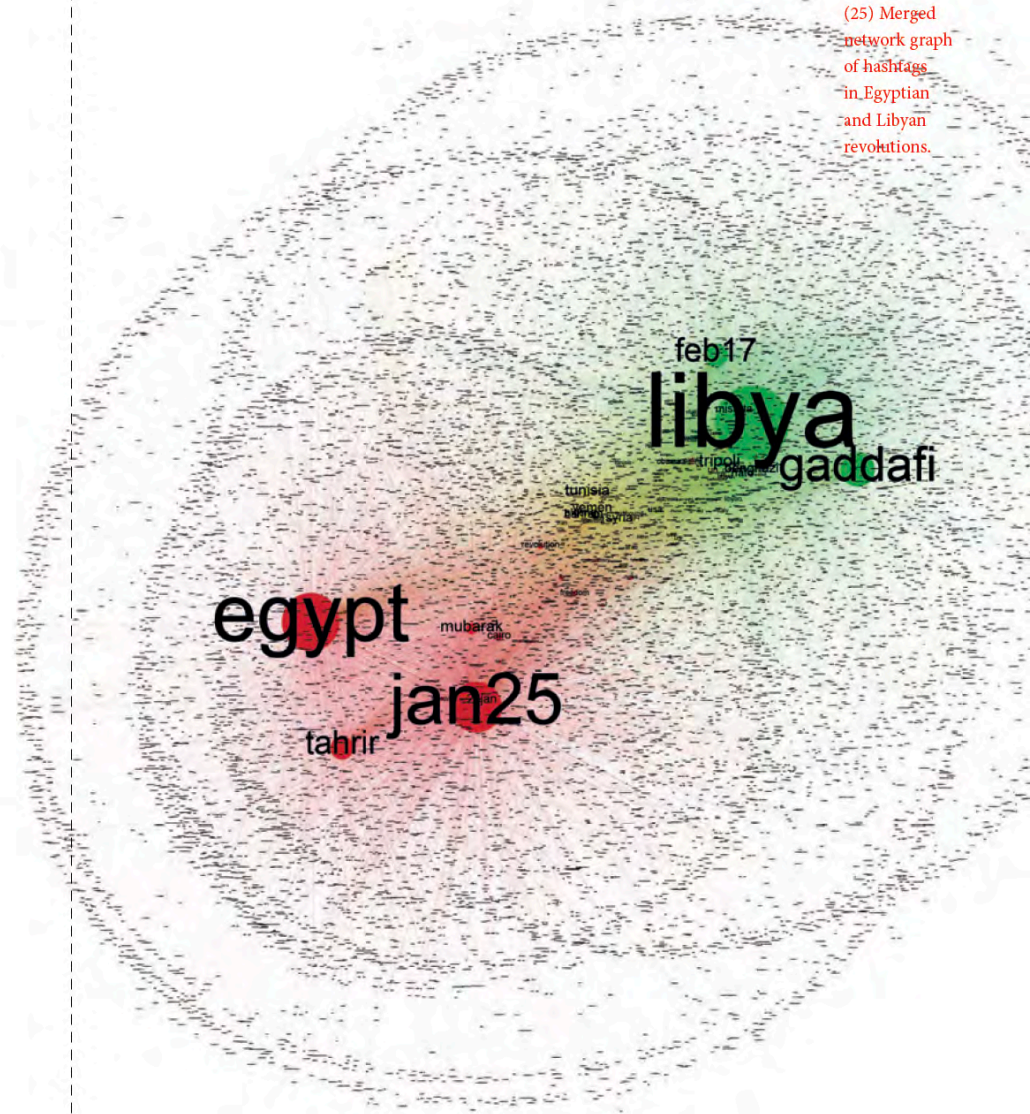


communities; most are just too small to see.

These solar systems/communities have their own centers, which highlight the many centers in a galaxy of hashtags. Derrida argues in "Structure, Sign and Play" that the center is a function not an intrinsic unity, a site of origin, or a telos.⁸³ What if the "function" of our center is a math function? In this case, it is the Louvain method for community detection, which discovers latent patterns in networks.⁸⁴ It shows that networks of speakers self-organize their communications around particular topics.

In Figure 22, we see yet another community, "other

(25) Merged
network graph
of hashtags
in Egyptian
and Libyan
revolutions.



@aroe_s, who has a Twitter space ded-

icated to information regarding earthquakes in Japan, tweeted more than 2,500 times during this time period (by far the most prolific user). The highest number of tweets per hour comes about a month after the earthquake: April 7 at 11:32pm. This is assuredly due to the occurrence of the second largest aftershock that shook Japan at magnitude 7.1 (there was actually a 7.9 earthquake that followed 30 minutes after the main 9.0 earthquake on March 11).

Sentiment Analysis

(YK) One aspect of social media is that its ephemeral nature can provide a snapshot of a moment's mood, reflected—in aggregate—by the content of what people are tweeting about in real time. The rise and fall of different emotions are a function of the many events that transpired in the aftermath: the evolving story around the nuclear meltdown and radiation contamination, the timing of the many devastating aftershocks, the increasing death toll, and the futility of finding survivors all contributed to the nation's psyche. In order to analyze the emotional and psychological state of the nation in the

199,741 (29%) distinct tweets (some of which are also retweets or duplicates with slight variations)



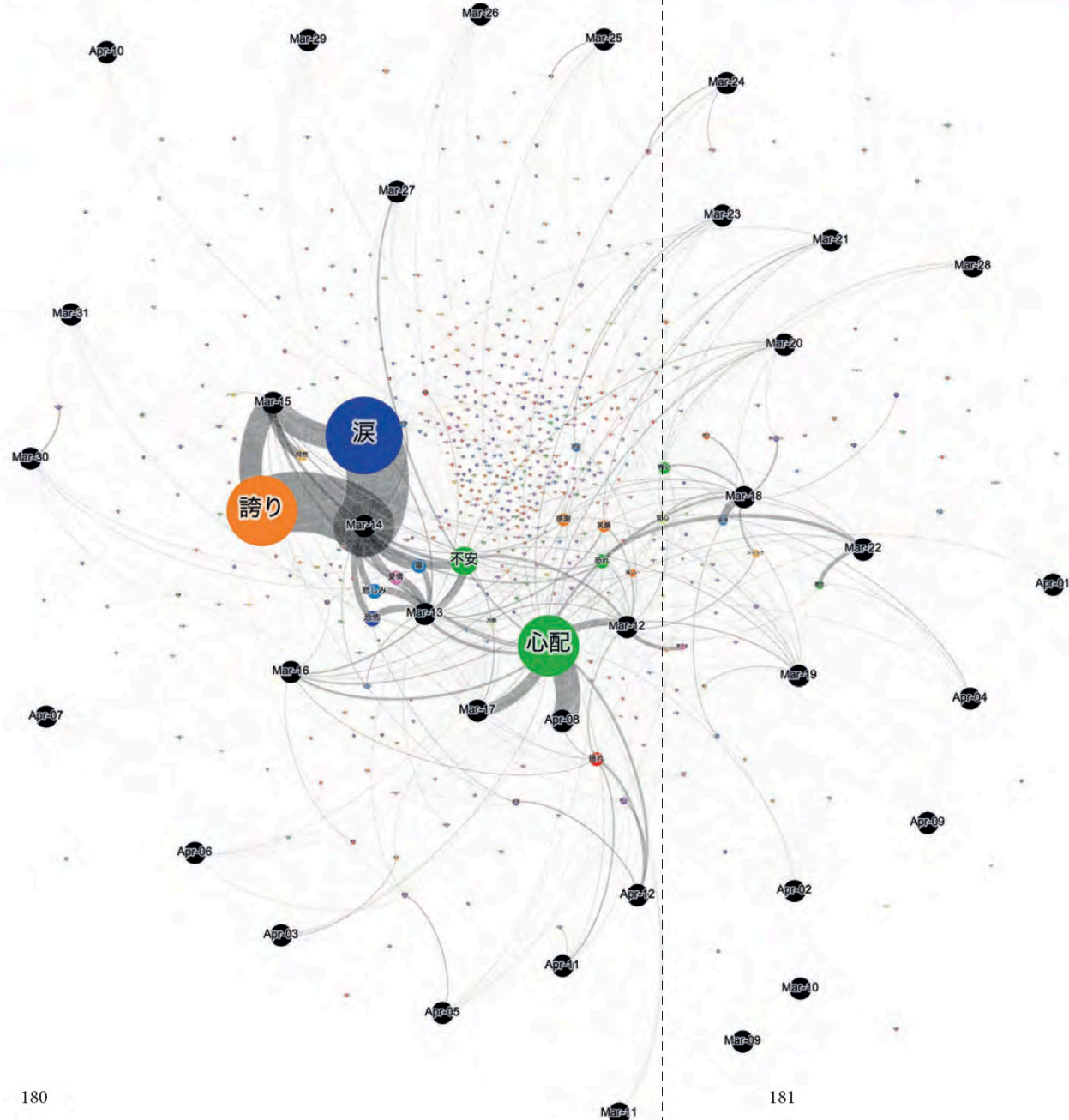
So what was the country “feeling” after the earthquake? Was it engulfed in sorrow? Anger? Fear? What effect did the hundreds of aftershocks have on the populace? Creating a universe represented by two object types—days (in black) and words (colored by emotion categories)—one can see the gravitational pull for the more “central” days and words. (Fig. 27) Unlike the multiple centralities that exist in the previous Egyptian universe, this one has a singular core, dominated by objects that orbit around a single, focused event. Words are sized by how often they were spoken, and their correlation to the different days is represented by lines that are also sized according to the usage per day. Communities are created by clusters of words, which are mentioned in proximity to each other. The two largest clusters are represented by the words 涙 (tears) and 誇り (proud), which have formed a giant galaxy around March 14 and 15. This reflects the content from @NamicoAoto’s single tweet (that was consequently retweeted more than 17,000 times) and embodies the nation’s anxious sentiments, days after the earthquake, expressing pride and sadness in the same context.

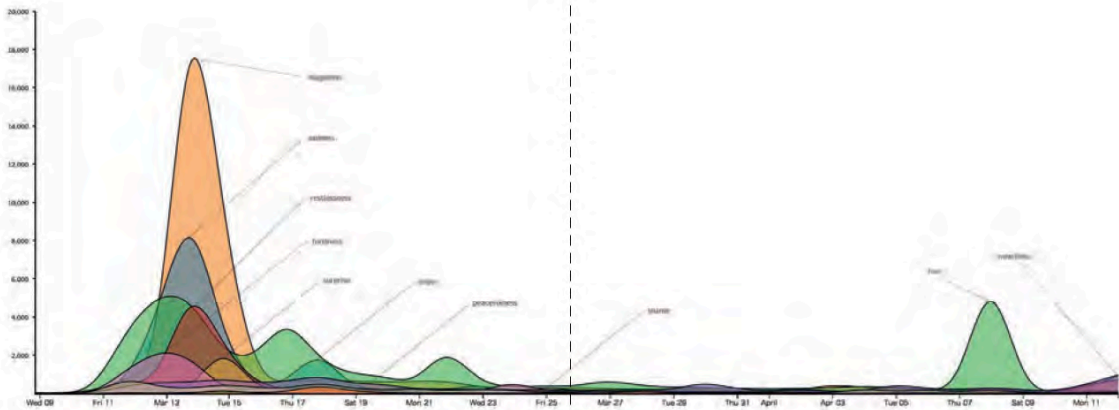
The next largest cluster is formed by the word 心配 (worry), a sentiment that is classified as “fearful,” and is marked by its presence following the two largest earthquakes, on March 11 and April 7. The final cluster community revolves around March 13, two days after the earthquake, showing strong connections between the words 悲しみ (sadness), 恐怖 (scared), 不安 (anxious), 愛情 (love), and 傷 (pain).

But what about replaying these emotions over time? The charts in **figures 28/29** reflect the 10 emotion categories over the 35 day period. Looking at the top chart, many of the sentiments are dormant during the first couple of days, but collectively explode shortly after. Perhaps this is because the initial “shock” of the event prompted content that was largely informational in nature. “Happy,” “sad,” and “restless” sentiments each peak three days later, on March 14 (“happy” peaks here because the word “pride,” from @NimicoAoto’s tweet, is classified in it). But “fear” does not reach its high point until April 8, the day after the second largest aftershock. The country experienced many earthquakes over the course of the month, averaging more than ten earthquakes a day. However, the number of earthquakes had tapered



(27) Network diagram showing emotion keywords that were prevalent after the earthquake.



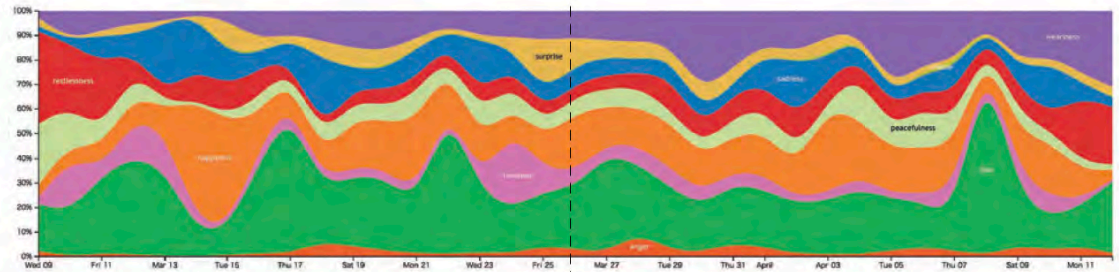


(28) Emotions
charted over time.

off considerably over time, perhaps explaining why the April 7 earthquake rekindled and renewed much of the fear in a nation that may have just started to begin its healing process.

While this chart effectively communicates the sheer quantity of emotional content, it does not reveal the distribution of the 10 emotions on a day-by-day basis and gives a false impression that the nation had no emotions between March 23 and April 7. To be sure, there was—measured in absolute numbers—significantly less emotional content in the Twitter archive during this period than in the days immediately

following the earthquake and tsunami. As such, the chart below was generated to better expose the significant differences between each emotion group, regardless of the total tweet count, as values were normalized and scaled to 100%. One can see that “fear” is a dominant sentiment throughout and “weariness” lingers and escalates with time. Both are expressions that bespeak an exhausted population dealing with the uncertainty of the on-going aftershocks and, at the time, the unbounded and uncontrolled nuclear catastrophe.

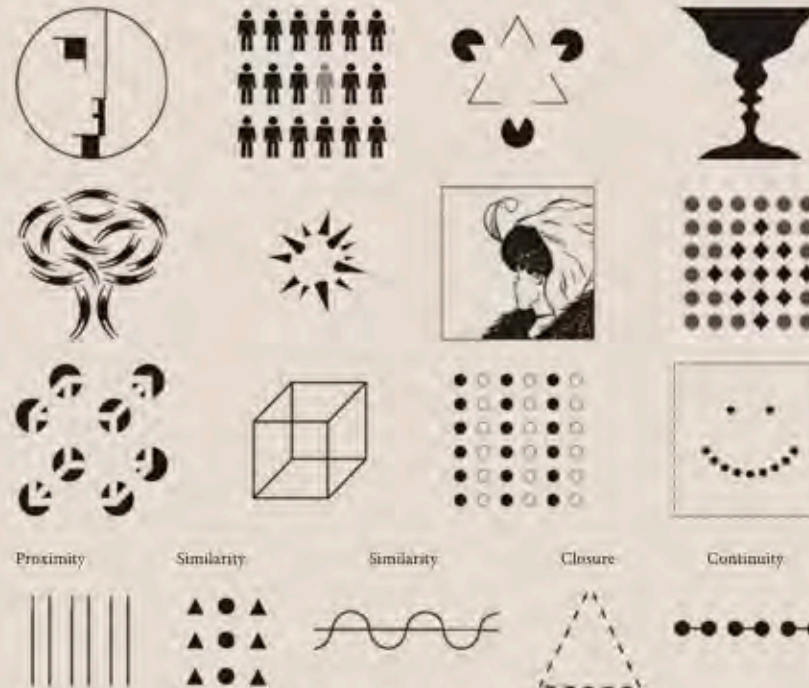


(29) Emotions
as relative percent-
ages over time.

Gestalt diagrams

The term “gestalt” refers to groupings and our tendency to see patterns wherever possible. Human perception isn’t literal. We will close gaps, see motion, make partial shapes into whole ones in ways that are surprisingly predictable. Biologists who study perception refer to the “ecology” of vision—ways our visual processes favor needs or tasks essential to our

survival. Such ideas counter the old “representational” approach to vision as a “picture in our heads,” and replace it with constructivist notions. We don’t simply *see* what *is* in a mechanistic way. Instead, *what is seen is what is made*. Instead of talking about pictures and images, we describe visual activity in terms of affordances and processes.



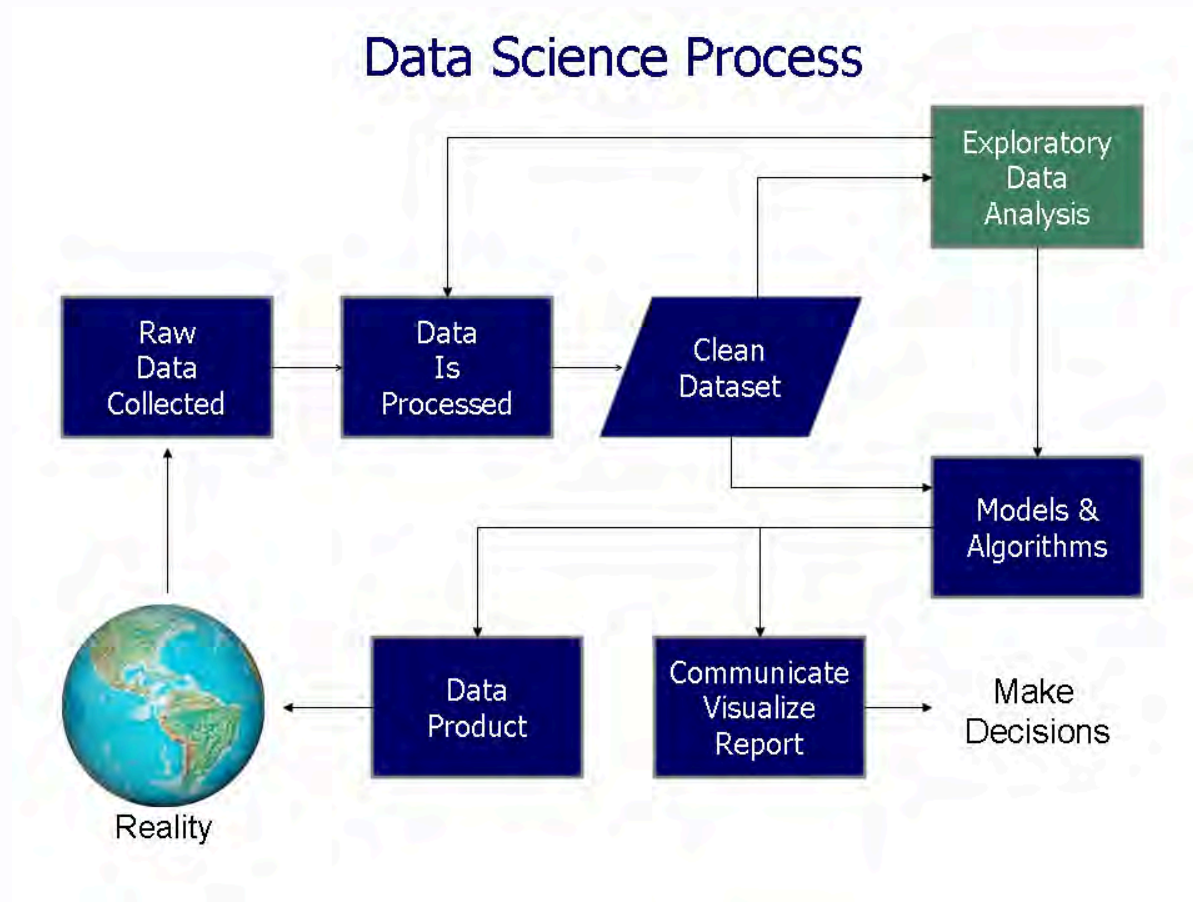
Gestalt diagrams, various sources.

**“The greatest value of a picture is when it
forces us to notice what we never expected
to see.”**



— John Tukey, 1977

EDA



Exploratory Data Analysis

① Discovery through Visualization

Seeing patterns, trends, and outliers that numbers alone cannot show.

可視化による発見

数字だけでは見えないパターンや傾向、外れ値を「見る」ことで発見する。

② Iterative Exploration

Analyze, visualize, question, and refine repeatedly to uncover deeper insights.

反復的な探究

可視化と分析を繰り返し行い、データの奥にある意味を掘り下げる。

③ Openness to Surprise

Expect the unexpected — EDA values curiosity over confirmation.

意外性への開放性

仮説の検証よりも、好奇心と「予想外の発見」を重視する。

`datasaurus.csv`



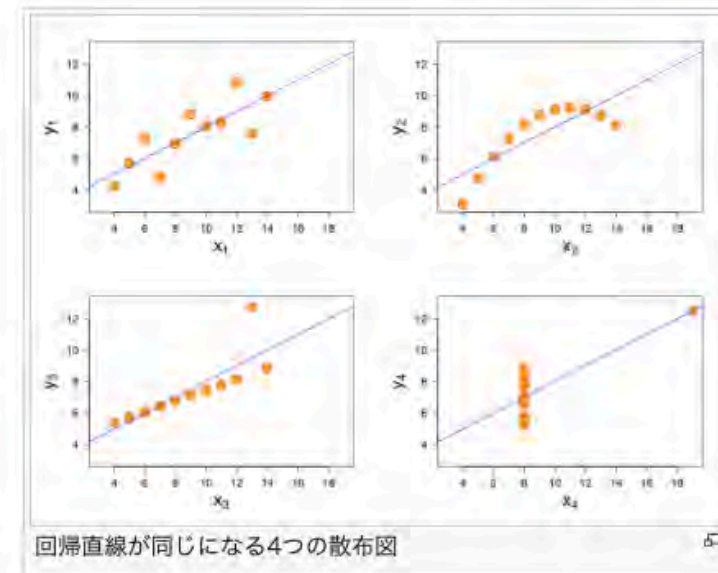
Find the dinosaur!

Anscombe's quartet

概要 [編集]

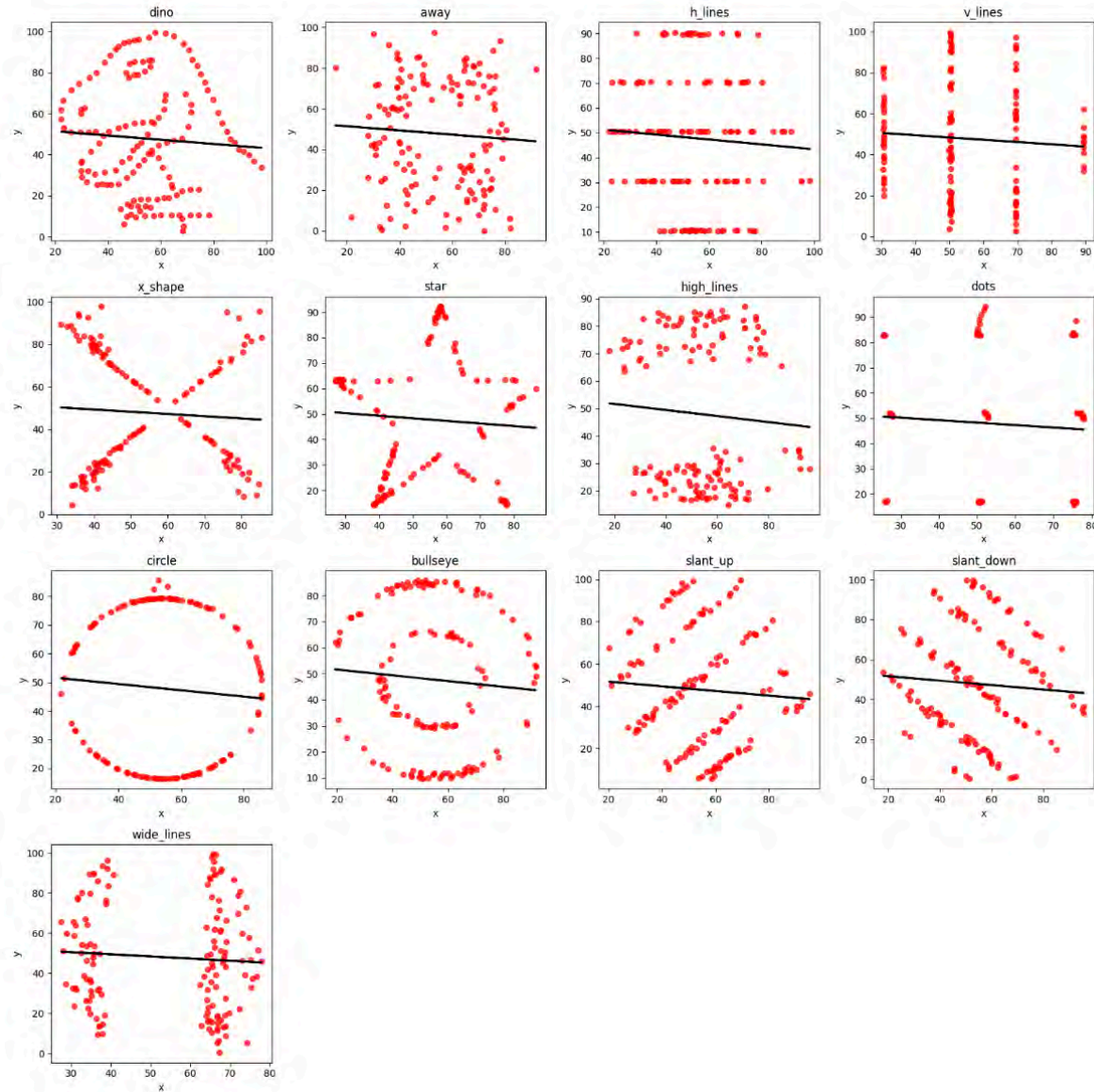
アンスコム の例は右のグラフに示された4つのデータセットからなる。4つ全てについて以下が成り立つ。

統計量	値
x の平均	9 (正確に一致)
x の標本分散	11 (正確に一致)
y の平均	7.50 (小数第2位まで一致)
y の標本分散	4.122 or 4.127 (小数第3位まで一致)
x と y の相関係数	0.816 (小数第3位まで一致)
回帰直線	$y = 3.00 + 0.500x$ (それぞれ小数第2位、小数第3位まで一致)



<https://ja.wikipedia.org/wiki/アンスコムの例>

In Python



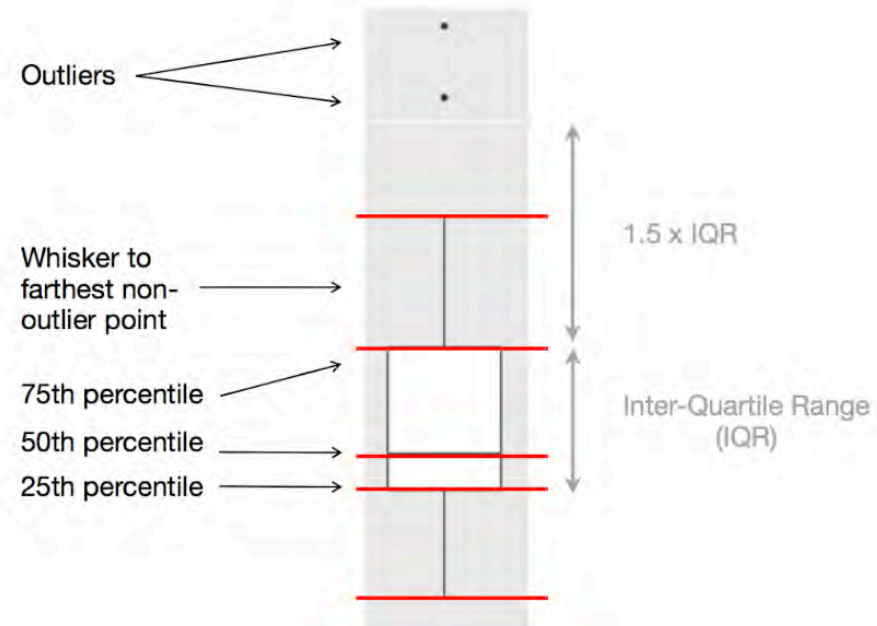
The actual values in a distribution

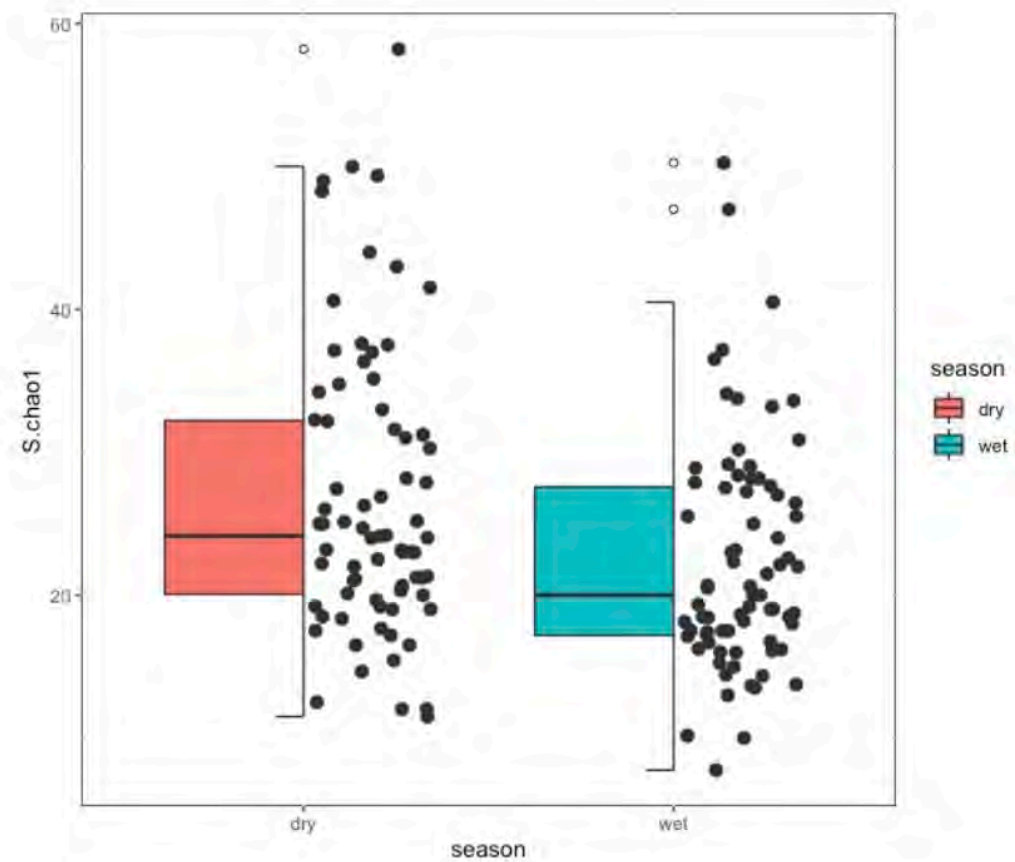


How a histogram would display the values (rotated)



How a boxplot would display the values

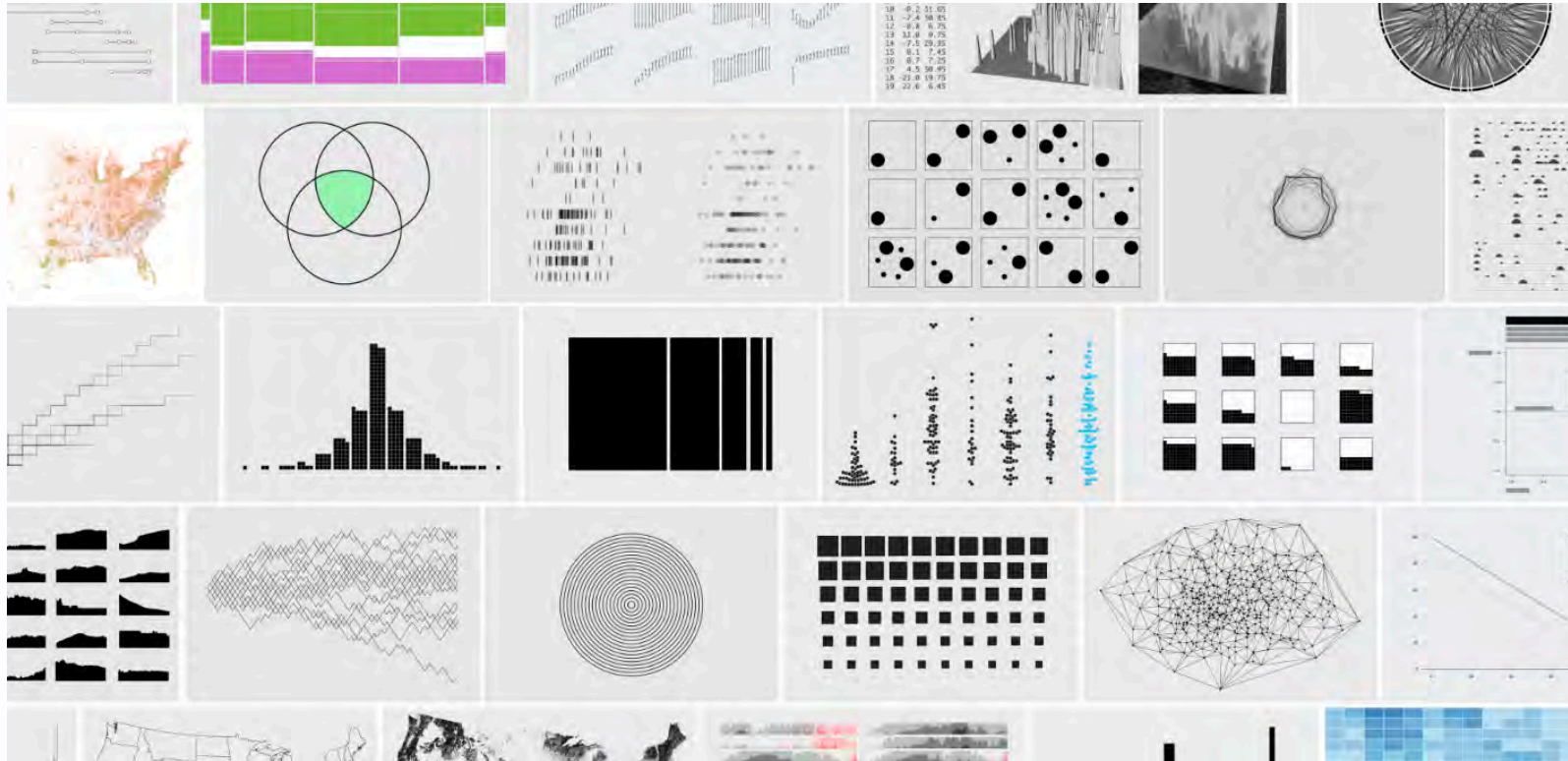




Visual Cues			
	Position	Length	Angle
Patterns			
Increase			
Decrease			
Combination			
Outlier			

<https://www.nature.com/articles/497186a>

Need inspiration?



<https://flowingdata.com/>

Week 4 課題